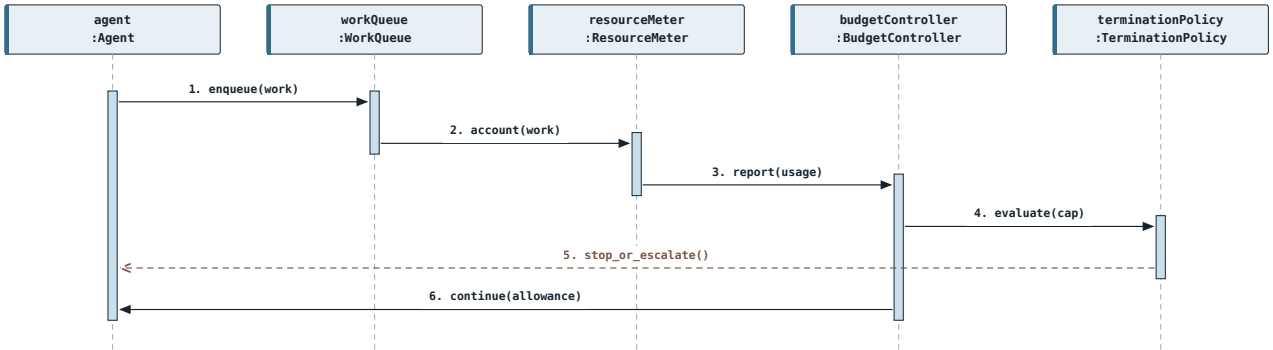
P19 Budgeted loop with caps

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Bound execution resources.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	The harness can consume bounded turns, steps, tokens, time, or cost.
TRIGGER	Accounting approaches or exceeds a configured cap.
STRUCTURE	Resource ledger, cap policy, terminal reason, and optional escalation.
EFFECT	Execution stops or changes policy before an unaccounted resource overrun.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Relevant resource use is observable and shared with delegated work.
ENSURES	Exhaustion produces a distinct terminal outcome and records the responsible cap.
INVARIANT	Budget exhaustion is never reported as task success.
VIOLATION	Missing accounting or handler failure falls back to a safe terminal policy.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.